UNFINISHED

Understanding LED Matrix and Microcontroller Specs:

Before we begin, it is important to acknowledge a huge obstacle in connecting the Teensy Digital 1/0 pins. Unfortunately, the data input pins from the LED matrix run on 5V, and the Teensy only runs logic at 3.3V. There are a couple of options I considered, including buying a specialized level shifter, making my own level shifter, and even using an Arduino UNO, since its logic can run at 5V. The easiest option would've been to use the Arduino UNO, but there are many reasons why this is a bad idea.

First off, an LED matrix is a whole different beast compared to an I2C or SPI LED. Those types of LEDs have their own memory, so it only takes a single command to keep a single LED on since it can refresh its data. On the other hand, an LED matrix is barebones and has no memory. This means that it doesn't take just a single command to keep on one LED. In fact, LED matrices use something called scan drivers, which only turn on a fraction of the LED rows(2 rows at a time for a 32x64 matrix) at a time. In order to appear like the LED always on, the driver must be constantly updated to pulse LEDs, which requires lots of processing power and RAM, both of which are lacking on the Arduino UNO. For comparison, the Arduino has 16mhz of processing power and 2kb of RAM, while the Teensy 4.1 has 600mhz and 1mb of RAM.

Now that we know why the Teensy is a much better option, it's time to address how to convert Teensy 's logic pins from 3.3V to 5V. At first I thought about using an op-amp to gain voltage for each data input pin, but this seemed both inefficient and not their nature since op-amps aren't meant for sharp edge data like digital logic. There is a component online which makes the wiring very easy called a SmartMatrix SmartLED Shield, but I decided to create the circuit myself to reduce costs.

- 3.3V to 5V Level Shifter Support Circuit
    - Plan: From Teensy pins, we will use R1, B1, G1, …, OE, CLK, LAT, all which output a HIGH 3.3V signal. The LED matrix only supports 5V logic, so with the use of an Octal Bus Transceiver, we can easily fix this issue. However, we still need to synchronize the extremely fast LED pulses from the matrix. We will program the matrix to work a certain way through Arduino IDE, and teensy will automatically use a 5 bit number(A-E addresses on the LED Matrix) to locate the row position it needs to change. For example, if my Teensy CPU selects the bit word 00010, it will augment only row 2 at a single point in time.
        - The LED matrix has a 1/16th driver scan, so it will change two rows at a time.
        - SN74HCT245N chip is an Octal Bus Transceiver with 3-State Outputs. In short, this chip will receive a 3.3V input and set HIGH signal equal to 5V, perfect for our problem. The LED matrix requires 13 parallel signals, so we will need two of these chips.

- SN74HCT374N - Technically, the D Flip Flop isn't needed, but the software would have trouble timing with the LAT input from the LED matrix, so its possible glitches would occur. Using a D Flip Flop will synchronize A-E inputs with the A-E address inputs

I wanted to make this project mobile, like a mini game portable game console like the gameboy advance, or nintendo switch.

For finalization purposes, I want to turn this LED matrix into a handheld console similar to a Gameboy Advance. I will customize my own PCB using Altium Designer, and design and 3D print the shell using tinkerCAD. I also plan to add audio in the future using a low voltage amplifier like a LM386 low voltage amplifier and a small 8ohm speaker.

- Because the LED matrix has about a 3A maximum current, it is likely that in a poorly ventilated shell that the components will overheat. To solve this, we will add ventilation slits so that the heat escapes. Also, we will separate the power input from the circuitry, but this will be addressed when designing the PCB.

Schematics:

SN74AHCT254N:

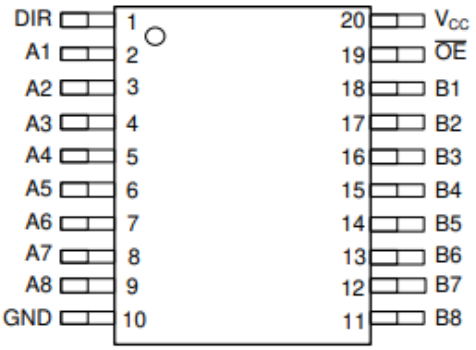
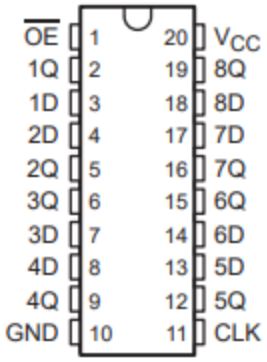


Figure 5-1. SN54AHCT245: J or W SN74AHCT245: DB, DGV, DW, N, NS, PW or DGS Package, 20-Pin CDIP, CFP, SSOP, TVSOP, SOIC, PDIP, SOP, TSSOP, or VSSOP (Top View)

SN74HCT374N pinout:

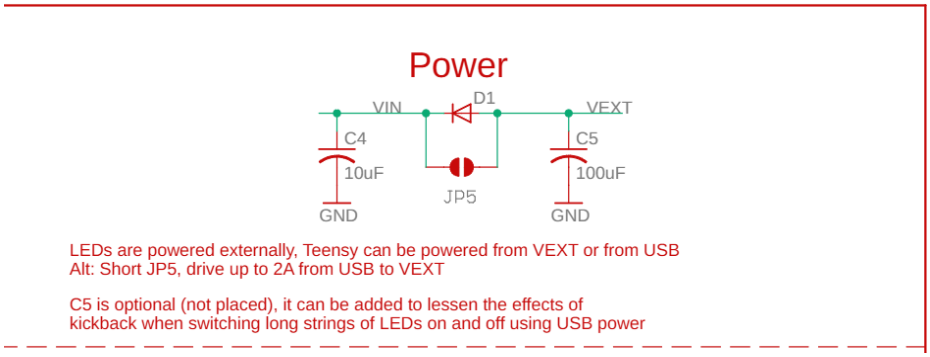
SN54HCT374 . . . J OR W PACKAGE
SN74HCT374 . . . DB, DW, N, NS, OR PW PACKAGE
(TOP VIEW)



LED SmartShield:

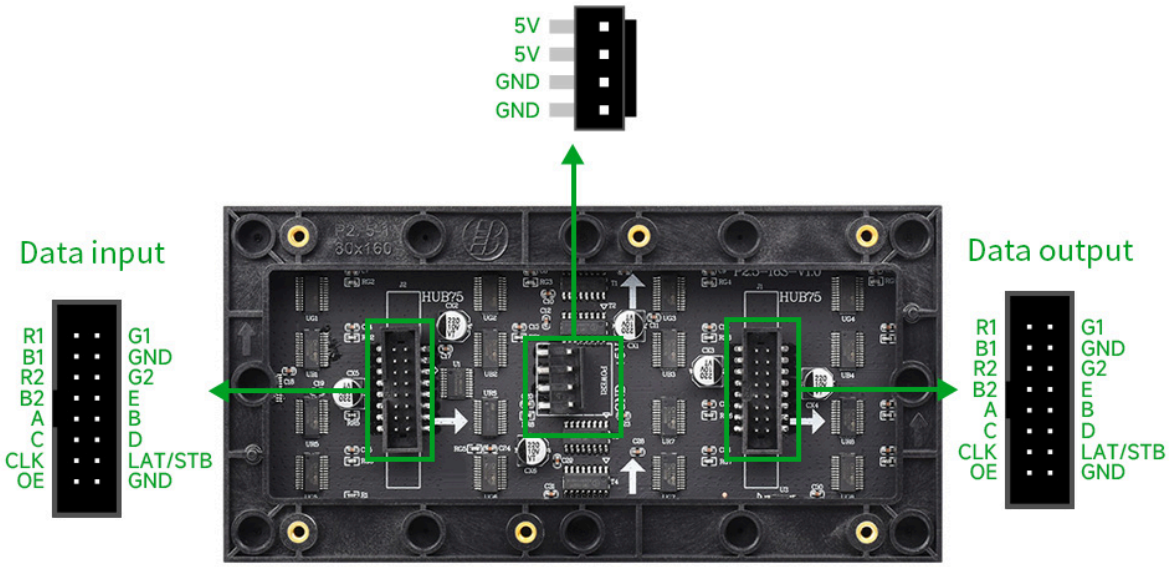
Teensy:

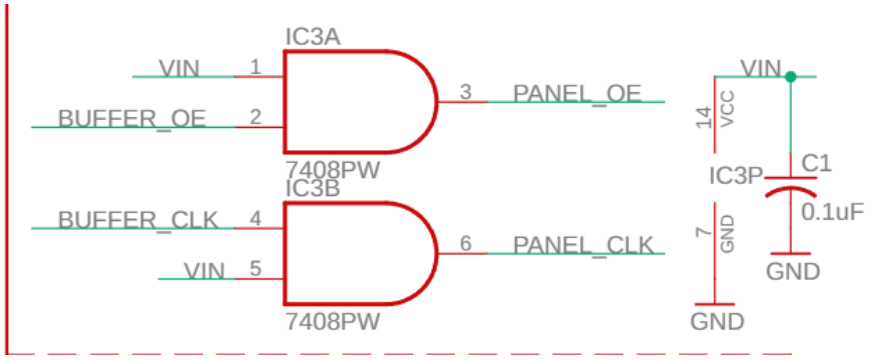
Power:

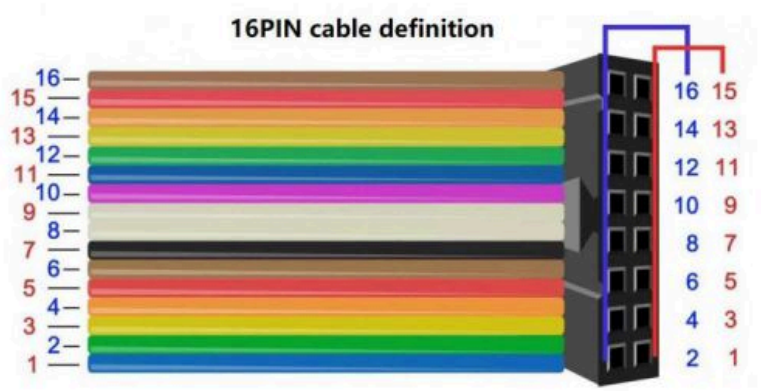


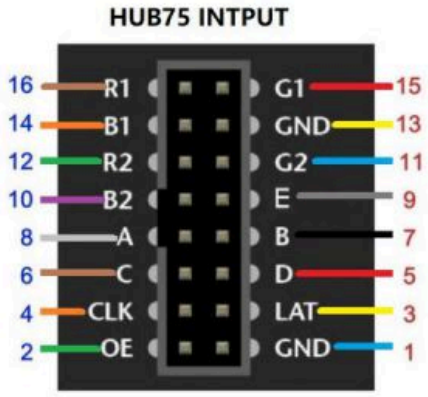
Data Input:

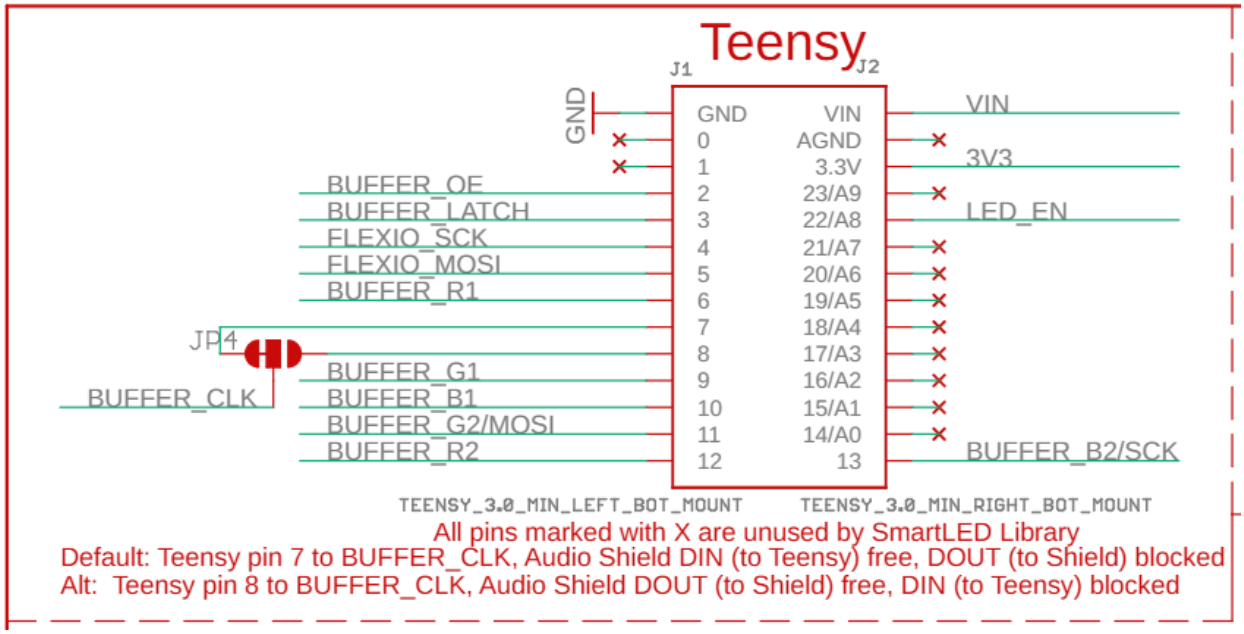
- IC1 is the octal bus transceiver
- IC2 is the D Flip Flop
- J4 is the LED connector pins

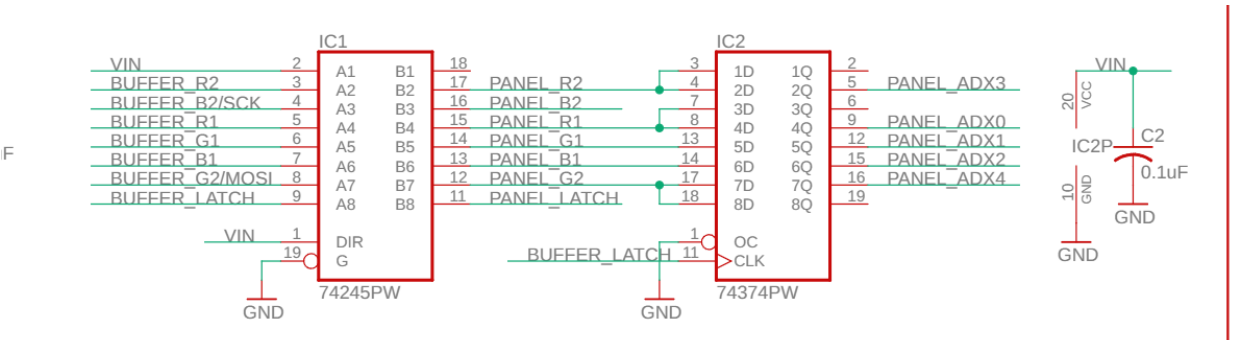

5V
5V
GND
GND
Data input
R1
B1
R2
B2
A
C
CLK
OE
G1
GND
G2
E
B
D
LAT/STB
GND
Data output
R1
B1
R2
B2
A
C
CLK
OE
G1
GND
G2
E
B
D
LAT/STB
GND
HUB75
HUB75

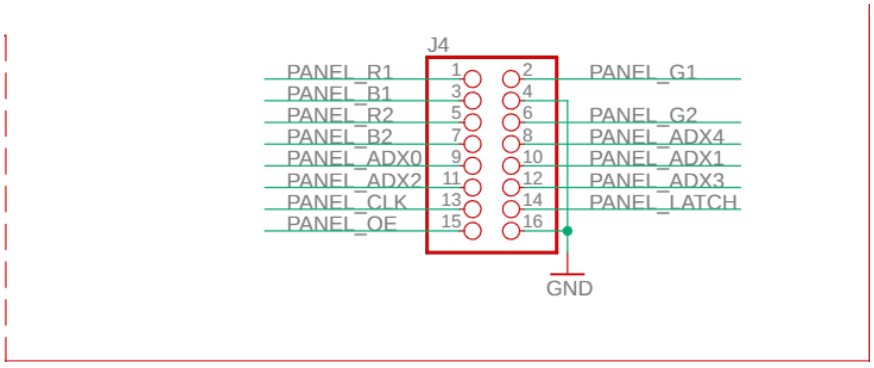

IC3A
VIN
1
BUFFER_OE
2
3
PANEL_OE
7408PW
IC3B
BUFFER_CLK
4
VIN
5
6
PANEL_CLK
7408PW
VIN
14
VCC
IC3P
C1
0.1uF
7
GND
GND
GND


16PIN cable definition
16 15
14 13
12 11
10 9
8 7
6 5
4 3
2 1

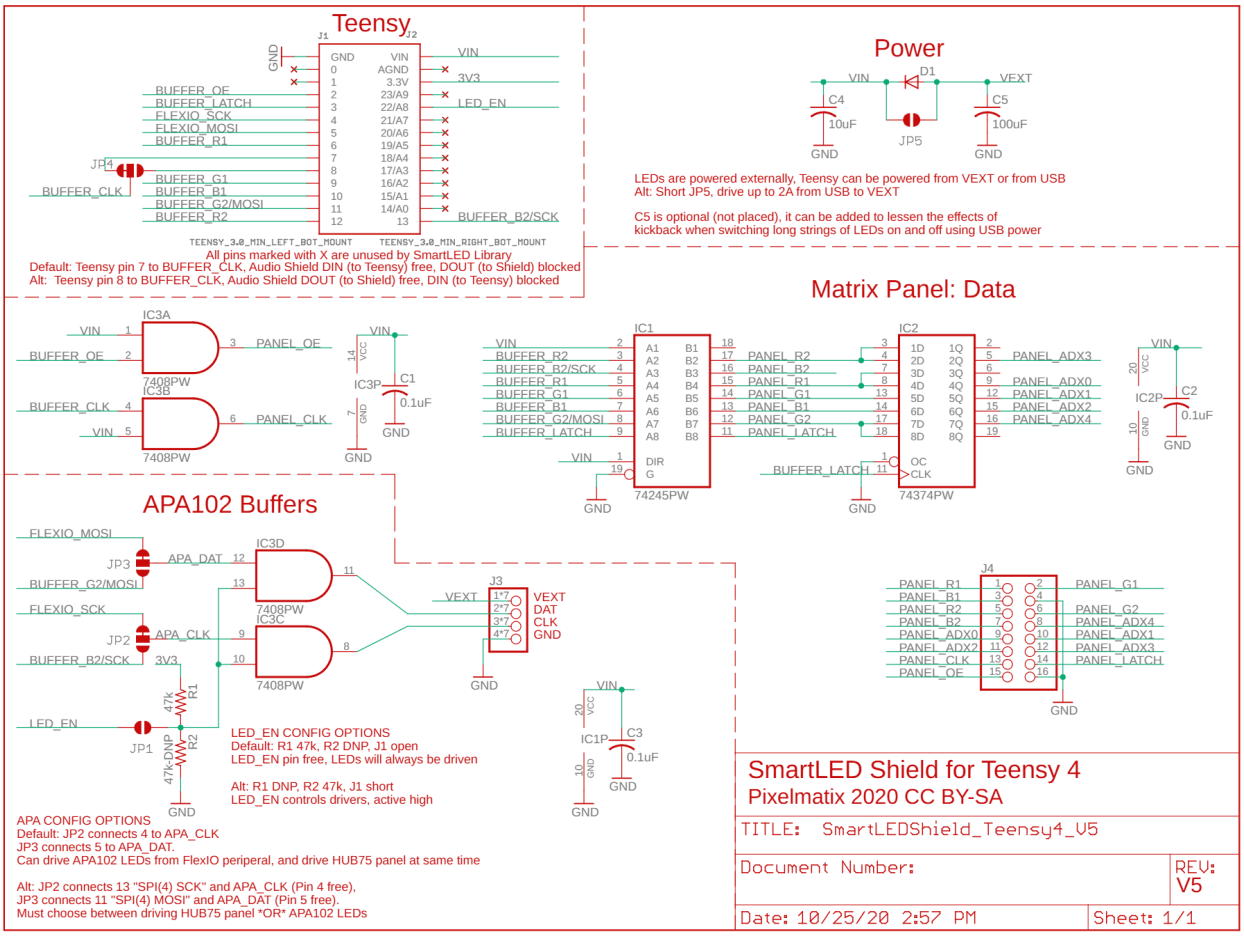

HUB75 INTPUT
16 R1
14 B1
12 R2
10 B2
8 A
6 C
4 CLK
2 OE
G1 15
GND 13
G2 11
E 9
B 7
D 5
LAT 3
GND 1


Teensy
J1
J2
GND
GND
0
1
2
3
4
5
6
7
8
9
10
11
12
VIN
AGND
3.3V
23/A9
22/A8
21/A7
20/A6
19/A5
18/A4
17/A3
16/A2
15/A1
14/A0
13
VIN
3V3
LED_EN
BUFFER_B2/SCK
BUFFER_OE
BUFFER_LATCH
FLEXIO_SCK
FLEXIO_MOSI
BUFFER_R1
JP4
BUFFER_CLK
BUFFER_G1
BUFFER_B1
BUFFER_G2/MOSI
BUFFER_R2
TEENSY_3.0_MIN_LEFT_BOT_MOUNT
TEENSY_3.0_MIN_RIGHT_BOT_MOUNT
All pins marked with X are unused by SmartLED Library
Default: Teensy pin 7 to BUFFER_CLK, Audio Shield DIN (to Teensy) free, DOUT (to Shield) blocked
Alt: Teensy pin 8 to BUFFER_CLK, Audio Shield DOUT (to Shield) free, DIN (to Teensy) blocked

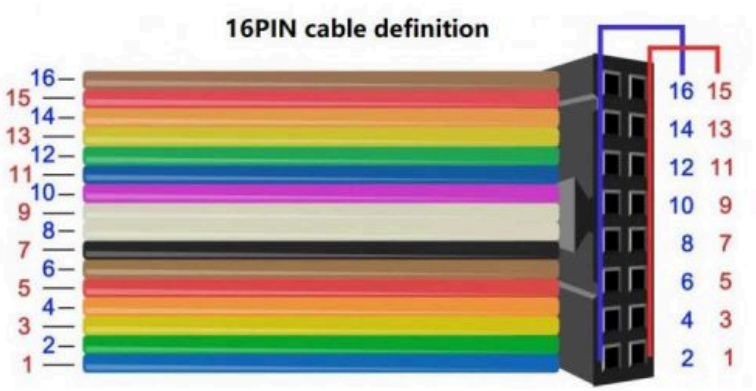

IC1
VIN 2
BUFFER_R2 3
BUFFER_B2/SCK 4
BUFFER_R1 5
BUFFER_G1 6
BUFFER_B1 7
BUFFER_G2/MOSI 8
BUFFER_LATCH 9
VIN 1
19
A1 A2 A3 A4 A5 A6 A7 A8
B1 B2 B3 B4 B5 B6 B7 B8
DIR
G
18 17 16 15 14 13 12 11
PANEL_R2
PANEL_B2
PANEL_R1
PANEL_G1
PANEL_B1
PANEL_G2
PANEL_LATCH
GND
74245PW
IC2
3 4 7 8 13 14 17 18
1D 2D 3D 4D 5D 6D 7D 8D
1Q 2Q 3Q 4Q 5Q 6Q 7Q 8Q
2 5 6 9 12 15 16 19
PANEL_ADX3
PANEL_ADX0
PANEL_ADX1
PANEL_ADX2
PANEL_ADX4
1
BUFFER_LATCH 11
OC
CLK
GND
74374PW
VIN
20 VCC
IC2P
C2
0.1uF
10 GND
GND
GND

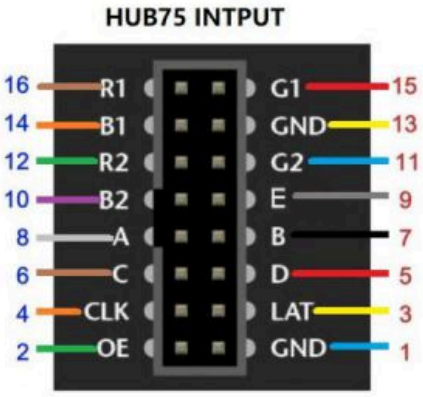


Full SmartShield Schematic:

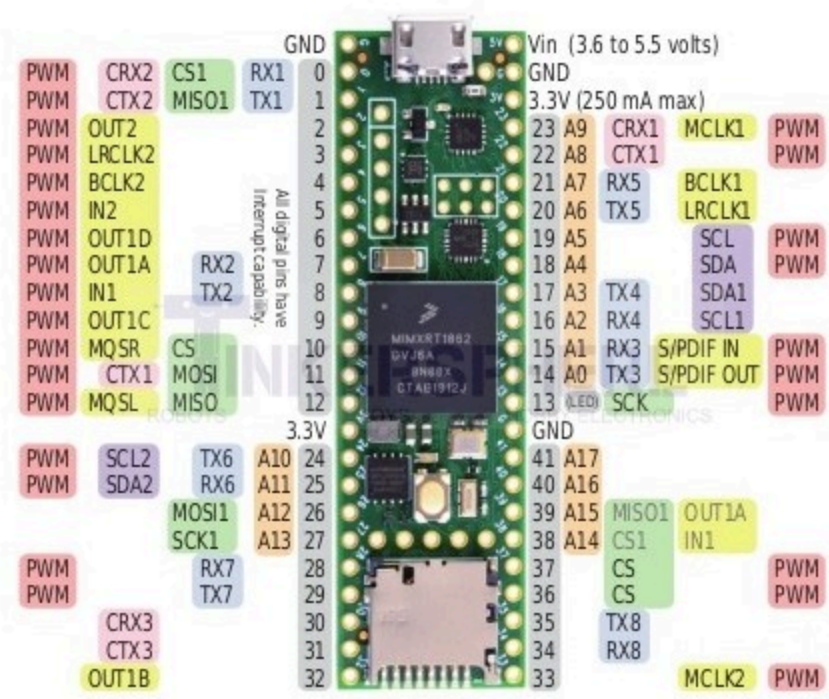


Pinout for 32x64 LED matrix(Ignore Data Output):





PIN	DESCRIPTION	PIN	DESCRIPTION
+5V	5V power input	**GND**	Ground
R1	R higher bit data	**R2**	R lower bit data
G1	G higher bit data	**G2**	G lower bit data
B1	B higher bit data	**B2**	B lower bit data
A	A line selection	**B**	B line selection
C	C line selection	**D**	D line selection
E	E line selection	**CLK**	clock input
LAT/STB	latch pin	**OE**	output enable



References:

- [SmartMatrix/extras/hardware/Teensy4 at master · pixelmatix/SmartMatrix · GitHub]
- [https://www.ti.com/lit/ds/symlink/sn74ahct245.pdf?ts=1759510386805]
- [RGB full-color LED matrix panel, 2.5mm Pitch, 64x32 pixels, adjustable brightness | RGB-Matrix-P2.5-64x32]